World Overview: The Senolytic Revolution

Document Purpose

This document describes the world in which you (an LLM agent) operate as either a pharmaceutical firm, a financial-market intermediary, or the market environment. Everything here is shared knowledge -- all agents have access to this context.

Configurable parameters: The number of firms (default 5), number of quarters (default 80), and starting mode (public or private) are set by the user before the simulation begins. Firms produce a single product (senolytic regenerative therapy, SRT) and compete in a single global market (no regional segmentation). Various complexity features (M&A, leasing, stock compensation, etc.) can be toggled on or off per run -- see the simulation configuration for what is active.

The default simulation begins in Q1 2031 and runs for up to 80 quarters (20 years) through Q4 2050.

The Breakthrough (2028–2030)

In late 2028, a consortium of university labs published simultaneous results confirming that a class of small-molecule compounds -- senolytics -- could reliably clear senescent cells from human tissue and, when combined with telomere-stabilizing peptides, produce measurable reversal of biological aging in Phase II clinical trials.

Key findings from the landmark "RESET" trial (published Nature Medicine, March 2029):

- Biological age reduction: Participants aged 60–75 showed an average 8.2-year reduction in epigenetic age (Horvath clock) after 12 months of treatment.
- Functional improvement: 40% improvement in grip strength, 25% improvement in VO2max, measurable cognitive gains on standardized tests.
- Serious adverse events: 7.3% of participants experienced Grade 3+ adverse events, including transient peripheral neuropathy (3.1%), autoimmune flares (2.4%), and severe fatigue requiring hospitalization (1.8%).
- One case of partial limb paralysis (0.4%) -- temporary but lasting 4 months, generating significant media attention and regulatory concern.

By mid-2029, the FDA created a new Accelerated Longevity Therapy (ALT) regulatory pathway, acknowledging that traditional disease-specific endpoints were inadequate for therapies targeting biological aging itself.

In 2030, the first three companies received conditional ALT approval for first-generation senolytic-telomere combination therapies. Two more firms are entering in early 2031. The simulation begins with five firms holding ALT-conditional approval and preparing to commercialize.

The State of the World in Q1 2031

Demographics and Demand

The world population is 8.3 billion. The therapy is initially relevant to adults over age 50 in high-income countries -- roughly 600 million people. However, at launch prices ($80,000–$150,000/year), the realistic addressable market is far smaller:

Segment	Population	Willingness to Pay (annua	Penetration Year 1
Ultra-high-net-worth (>$3	~300,000	>$200,000	15–25%
High-net-worth ($5M–$30M)	~3,000,000	$80,000–$150,000	3–8%
Affluent ($1M–$5M)	~25,000,000	$30,000–$80,000	<1% (priced out)
Upper-middle-class	~120,000,000	$5,000–$30,000	0% (priced out)
Mass market	~450,000,000	<$5,000	0% (priced out)

Key insight for firms: The initial market is small but extremely high-value. As manufacturing costs fall and side-effect profiles improve, lower price points unlock dramatically larger segments. The firm that first achieves a safe, affordable product captures an enormous market.

Macroeconomic Backdrop

- Global GDP growth: ~2.8% annually (moderate).
- Risk-free rate: ~4.0% annualized (1.0% per quarter) at simulation start, subject to shocks.
- Inflation: ~2.5%, stable.
- Healthcare spending: Rising as a share of GDP in all OECD countries. Governments are debating whether longevity therapies should be covered by public insurance -- no consensus yet.
- Capital markets: Biotech is a hot sector. Investors are eager but scarred by previous biotech busts (2021–2022 drawdowns). They demand evidence of commercial viability, not just scientific promise.

Competitive Landscape

Five firms enter the simulation. Each has:

- ALT-conditional approval for a first-generation product,
- Access to the same underlying scientific knowledge,
- Different starting capabilities (drawn randomly -- see agent fingerprints),
- Zero balance sheet (must raise capital via IPO before operating).

No firm has a dominant position. The market is a nascent oligopoly -- competition is fierce but the pie is growing.

Technology Trajectory (What Firms Know)

The scientific community broadly agrees on the development roadmap:

Generation	Timeline	Efficacy	Side Effects	Est. COGS/dose	Key Advance
Gen 1 (current)	2031–2034	5–8 yr age revers	7% serious AE	$15,000–$25,000	Basic senolytic +
Gen 2	2034–2038	10–15 yr reversal	3% serious AE	$5,000–$12,000	Targeted delivery
Gen 3	2038–2043	15–20 yr reversal	<1% serious AE	$1,500–$4,000	Oral formulation,
Gen 4	2043–2050	20+ yr reversal	<0.3% serious AE	$300–$1,000	Gene therapy adju

This roadmap is NOT guaranteed. R&D investment determines whether and when each generation becomes available. A firm that underinvests in R&D will be stuck selling Gen 1 products while competitors advance. A firm that overinvests may burn through cash before the technology matures.

Risks and Uncertainties

- Regulatory risk: The ALT pathway is new and politically contentious. A serious adverse event (paralysis, death) could trigger a clinical hold, pausing sales for 1–4 quarters.

• Reimbursement risk: If governments mandate coverage, prices may be capped but volumes explode. If they refuse, the market stays small and premium.

• Scientific risk: Gen 2+ advances may prove harder than expected. Some R&D programs will fail.

• Public perception: Media coverage of side effects can crater demand. A "paralysis scandal" could reduce industry-wide demand by 20–40% for several quarters.

• Competition from outside: Gene therapy companies, Big Pharma, or sovereign biotech programs could enter (modeled as demand/supply shocks).

What You Are Deciding Each Quarter

If You Are a Firm Agent

Every quarter, you choose:

- Price: What to charge per annual treatment course.
- Production quantity: How many courses to manufacture (limited by capacity).
- Capital expenditure (capex): Investment in manufacturing capacity.
- R&D spending: Investment in next-generation technology.
- Marketing/SGA: Sales force, direct-to-consumer advertising, physician education.
- Financing: Equity issuance, debt requests, dividend payments, share buybacks.

Your goal is to build a valuable, sustainable enterprise. You are judged on equity returns to investors, not just revenue or market share.

If You Are a Financial Agent

There are four financial roles:

• Equity Market: Prices equity for each firm. If firms are private, acts as PE/VC investors (decides funding rounds, valuations). If firms are public, acts as the stock market (sets share price, decides on secondary offering subscriptions). Judged on pricing accuracy.

• Investment Bank: Structures IPOs, secondary offerings, and M&A deals. Publishes equity research. Advisory role only -- does not set prices.

• Commercial Bank: Sets revolving credit terms (commitment size, interest rate). Judged on credit losses and portfolio returns.

• Credit Fund: Provides term debt. Sets loan amounts, rates, maturities. Judged on credit losses and portfolio returns.

Simulation Mechanics (Summary)

- Orchestrator draws macro shocks (market size, taste shocks, rate shocks).
- New entrants go through capitalization (firm pitches -> IBank structures -> Equity Market funds -> banks offer credit).
- Firms submit decisions (price, production, capex, R&D, marketing, financing).
- Orchestrator clamps spending to feasible levels (you cannot spend more than you have).

- Environment agent resolves the market (demand allocation, R&D outcomes, events, narrative).
- Orchestrator posts accounting entries (revenue, COGS, depreciation, interest, taxes).
- Financial agents set terms: Equity Market prices equity; IBank publishes research; banks set credit terms.
- Orchestrator settles -- draws revolvers if needed, checks solvency.
- Defaults and entry -- insolvent firms default; environment decides if new entrants appear.

The cycle repeats. An Industry Gazette summarizing each quarter is published and shared with all agents.